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MSC Apex

What's new in 2025.1

February 2024

A modern, direct modeller and mesher with full coverage of MSC Nastran for linear, nonlinear and topology optimisation workflows and direct access to cloud compute

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New capabilities and updates

MSC Apex 2025.1

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Release Highlights

New in MSC Apex in 2025.1:

MSC Nastran beam consistency

- Use the same ID across parts and remain consistent with MSC Nastran.
- Model joints more accurately and easily orient beams.
- Organise beam elements easily in pre- or post-processing.

Improving MSC Nastran and MSC Apex integration

- Apply 3D solid composite element properties or gasket materials to meshes for better accuracy and mesh quality.
- Solve collapsed elements and control how elements are treated for easier post-processing.
- Directly obtain Root Mean Square (RMS) Von-Mises stresses during random response analyses to evaluate structural performance under random loading.
- Improve convergence with a new element artificial damping energy case-control command.
- MSC Apex now uses the new high-performance nonlinear solver by default, which offers faster convergence and provides faster results.
- Choose your preferred composite strain output.
- Composite models are easier to handle, with new selection tools for zones and panels that don’t interfere with geometry or mesh picking.
- Perform transient analysis in MSC Apex using the full capabilities of SOL 400.
- Try new options when creating new scenarios to automatically create new scenarios based on default settings or run custom scripts that can define any custom settings.

Geometry and meshing enhancements

- Update node coordinates easily and get improved mesh quality.
- Mesh controls and seed points offer advanced hex meshing capabilities.
- Skinny surface meshes and proximity meshes result in fewer triangles and better mesh quality.
- Get more control over mesh quality when creating tri elements using the all-quad element option.
- Modify solid geometry while retaining midsurfacing capabilities.
- Easily visualise and manipulate sections of models using cutting plane functionality directly in pre-processing.

Post-processing improvements

- Create section force sensors directly during post-processing.
- Easily isolate entities for post-processing with new plot targeting for groups and visible elements/nodes.
- Execute custom-built utilities in post-processing.
- Dynamic analyses in post-processing now include a vibration view for more detailed results.
- Import scenarios and models directly from H5 files.

Infrastructure improvements and performance

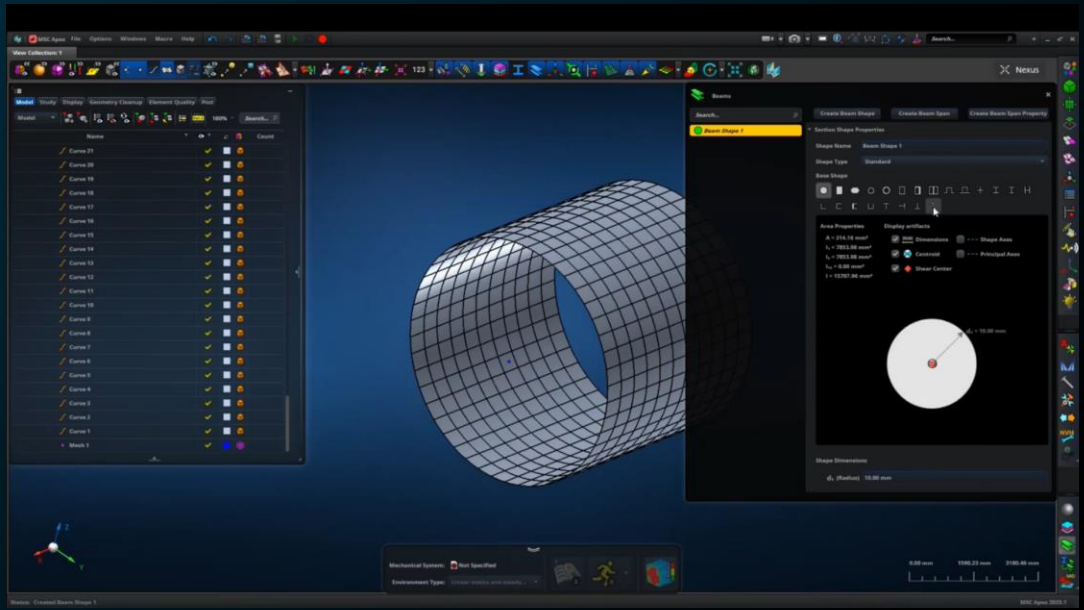
- Calculate generative behaviours for edge ties three times faster.
- Complete STL imports four times faster.
- Solve models with large numbers of nonstructural masses much quicker.

New capabilities and enhancements in MSC Apex 2025.1

MSC Nastran modeller | beams enhancements

Description:

The beam enhancements in MSC Apex 2025.1 is the first part of a comprehensive project to deliver robust beam functionality, mirroring the successful approach implemented for composites. This initial phase prioritises full MSC Nastran support and improved usability for beam-related workflows. A significant advancement is the introduction of a new beam property object, which enables the use of the same property ID across multiple parts and bodies. This design choice is expected to significantly reduce the number of property cards in exported BDF files and allow a single property to span multiple discontinuous beam segments across various parts.



Benefits:

- Use the same ID across multiple parts and bodies to maintain consistency with MSC Nastran properties and IDs with the new beam span property object.
- Model joints more accurately with new pin flags, and easily orient beams with G0 and offsets relative to the shear centre.
- Organise beam elements more easily in pre- or post-processing with beam spans, which also support mesh-only models.

Who needs it / Use cases:

- Companies using beam elements in modelling; typically, users in the aerospace and shipbuilding industries

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New capabilities and enhancements in MSC Apex 2025.1

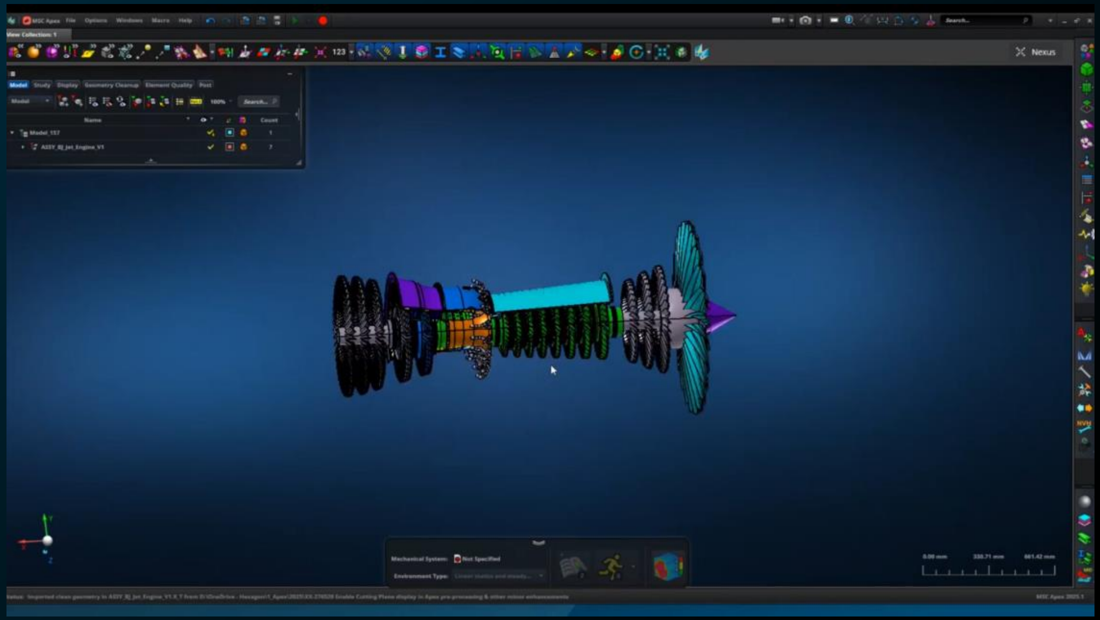
MSC Nastran modeller | section view

Description:

MSC Apex 2025.1 introduces long-awaited cutting plane functionality directly into pre-processing, building upon its existing capabilities in post-processing. The cutting plane's location and orientation are also retained across pre- and post-processing sessions, ensuring a coherent workflow.

Benefits:

- This enhancement allows users to easily visualise and manipulate sections of their models by showing clipped views or slices.
- Extensive controls for adjusting location and orientation via the button and toolbar, consistent with post-processing. A key benefit is that it caps both solid geometry bodies and meshes when cut.
- Both solid geometry bodies and meshes when cut, providing clearer structural inspection.



Who needs it / Use cases:

- All customers using MSC Apex to visualize solid or complex shell models

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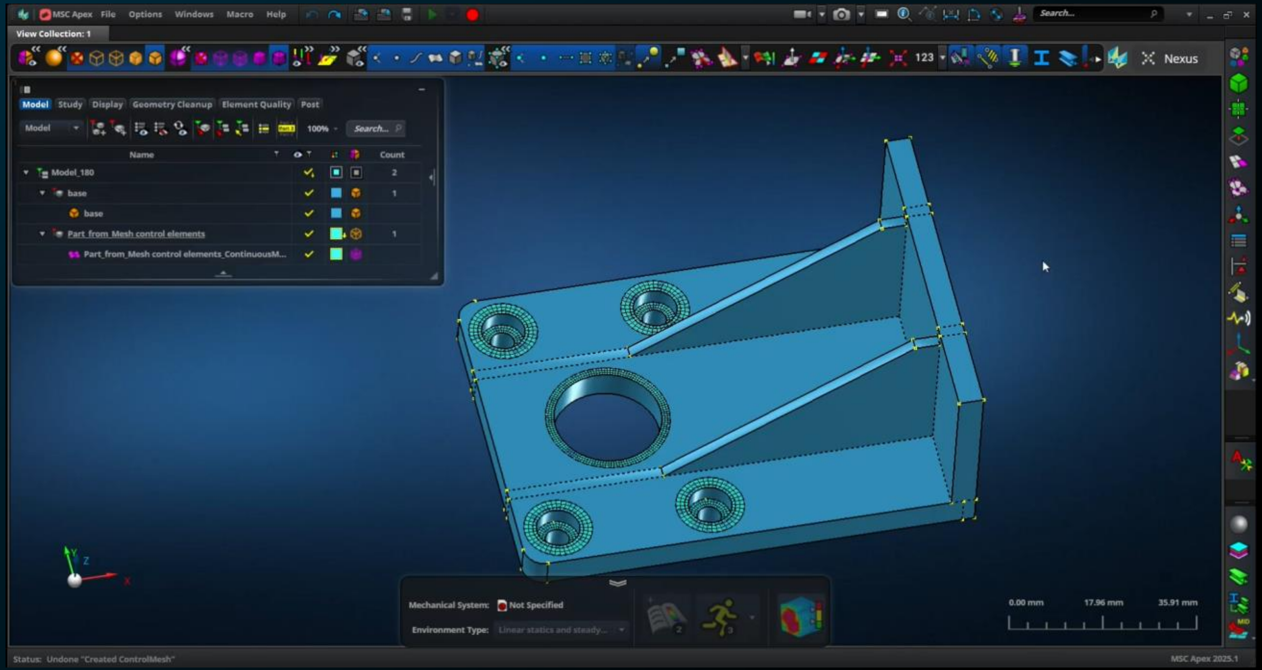
New capabilities and enhancements in MSC Apex 2025.1

MSC Nastran modeller | geometry and meshing

Description:

MSC Apex 2025.1 introduces notable advancements in geometry and meshing, primarily aimed at improving mesh quality and offering enhanced control during mesh generation. Key updates include a mesh improvement tool to update node coordinates for better element quality, new hex meshing capabilities with control objects, and refined handling of skinny and proximity meshes. Additionally, users gain more control during tri-element creation and enjoy improved solid geometry manipulation features, such as midsurfacing with cell faces.



Benefits:

- Mesh control objects for hex meshing automatically propagate and maintain a hex meshable solid, giving greater control and better-quality elements.
- Achieve triangle elements and better quality near boundary edges.
- Overall element quality improvements enhance the mesh improvement tool.

Who needs it / Use cases:

- Aerospace or automotive companies requiring strict controls over the mesh quality and local mesh features

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New capabilities and enhancements in MSC Apex 2025.1

MSC Nastran modeller | post-processing

Description:

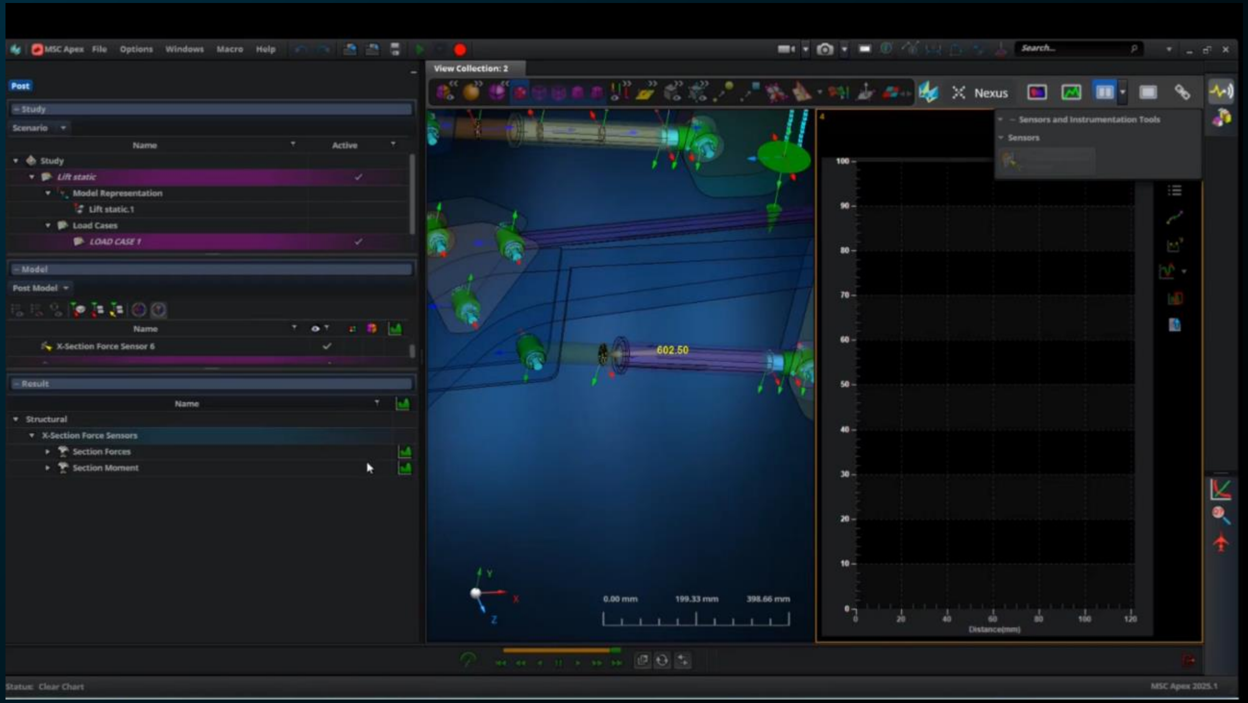
MSC Apex 2025.1 enhances section force sensors, allowing them to be created directly within post-processing. Previously, these were mainly used for Solution 101, but now sensors created in pre-process are shareable across scenarios, while post-process sensors are tied to the specific scenario. This provides greater flexibility in workflow, enabling users to manage and analyse sensor data more directly, including copying data from the data grid and setting sensors as plot targets for detailed review.

Benefits:

- Quickly hide/show entities in post-processing.
- Create cross-section force sensors without having to go back into pre-processing or re-run the analysis.
- Easily incoporate custom-built post-processing utilities.

Who needs it / Use cases:

- All engineers performing post-processing



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Previous releases

Highlights from MSC Apex in 2024

>> MSC Nastran case control and solution sequences

Comprehensive support for MSC Nastran adding many more options for SOLs 101, 103, 105, 108, 109, 111, 112, and 400.

>> New Capbilities for MSC Nastran Dynamics

Both pre- and post-processing support for frequency, transient, shock, and random response.

>> Mesh Control Regions

Freeze/lock mesh and associate it to geometry allowing for fixed mesh patterns near fasteners, welds, etc.

>> User Defined Graphics

Users can now customise their workflows without any restrictions while leveraging Python, the most widely utilised open-source language.

>> Run MSC Nastran through Nexus Compute

A new Nexus button now appears in the MSC Apex interface, allowing users to quickly connect to the cloud. Upon connection, compute requirements can be specified within the study tab while creating an input MSC Nastran job deck, following the usual process.

>> Generative Design Mission Switch

A model created through generative design can be validated with MSC Nastran by switching between the MSC Nastran and MSC Apex Generative design missions without closing the application.

>> Improved Composites

Construct composite models using either the element property or layered panels workflows with enhancements to managing generative stacks, global ply IDs, and both 2D and 3D composite properties.

Highlights from MSC Apex in 2023

>> **Shrink wrap meshing and improved hybrid mesh**

Shrink-wrap meshing quickly combines multiple geometries in an assembly to create a single continuous shell mesh.

>> **MSC Nastran loads and boundary conditions**

This release added extended support for loads like rotational force, load combination, and distributed loads. Additional constraints like SUPPORT and new initial conditions.

>> **MSC Nastran parameters and system cells**

All MSC Nastran parameters and system cells, including the creation of sets that can be selected during analysis, are supported.

>> **MSC Nastran output requests**

All MSC Nastran output requests, including the ability to create output on groups, are supported.

>> **Preprocessing support for MSC Nastran Modules**

Use the new MSC Apex modules mission to create modules for MSC Nastran. Translate, mirror, or connect your models and the instructions are written to MSC Nastran without creating duplicate nodes or elements.

>> **MSC Nastran H5 attach**

Increase performance by attaching results instead of importing them into the model. This update also includes additional support for MSC Nastran output in existing solution sequences.

>> **Many new community tools**

The MSC Apex scripting community continues to deliver new tools to rapidly extend pre/post capabilities. New tools include stress linearization, drop test automation, support for MSC Nastran Smart Super Elements, and many more.

Highlights from MSC Apex in 2022

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ID management

Control solids, shells, and beams.

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Materials

Many new MSC Nastran materials are supported.

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Properties

Many new MSC Nastran properties are supported. 2D element properties are enhanced. Convert layered panels to facet geometry. Discrete connector enhancements are included.

>>

Collaboration and modules

Visualise and post-process MSC Nastran modules through a new mission in MSC Apex.

>>

Fasteners (CFAST)

Open the generative design database in MSC Apex. With the load transfer to the design candidate option, only the mesh needs to be created to run a simulation in MSC Apex with the same model setup.

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Integrated MSC Nastran user experience

Connect MSC Apex to the external MSC Nastran solver. Migrate the integrated MSC Apex Structures solver to MSC Nastran.

>>

Meshing enhancements

Visualisation and post-processing support were added for MSC Nastran hybrid (pyramid) elements. Additional shell and solid meshing enhancements include improved through-thickness controls on solids.

Highlights from MSC Apex in 2021

Adams modeller

This was the first release based on the MSC Apex framework with integration to the Adams solver to support rigid- and flex-body dynamics.

Groups

Create and visualise entities based on their group assignments as well as import and export MSC Nastran through the SET3 entry.

Composites enhancements

MSC Nastran PCOMP and non-manifold geometry are supported.

Collaboration tools

Import one MSC Apex database into another and use MSC Nastran INCLUDE files.

MSC Nastran nonlinear analysis

MSC Nastran SOL 400 including contact (both node-to-segment and segment-to-segment), elasto-plasto materials, and post-processing of nonlinear results are supported.

Lug load and traction load

Automatically generate loads to simulate lug load conditions and support for pressure loads applied in any direction, including total load.

Cutting planes in post-processing

Slice the model to visualise the results along a cut view

Highlights from MSC Apex in 2020

>> Generative design

Apex Generative Design integrates AMendate Generative Design topology optimization software for additive manufacturing.

>> New light theme

New theme provides more options for screenshots, videos, and other reporting tools.

>> Exploded view

Visualise and update connections between parts in an assembly.

>> MSC Apex custom utilities

Custom tools can be internally managed through GitHub and over 100 utilities are automatically shipped with MSC Apex.

>> Thermal expansion

New support for temperature loads allows running thermal expansion analysis with MSC Nastran.

>> Pressure and temperature mapping

Import data from a table or directly from CFD to visualise it in MSC Apex. Data will be mapped from the existing mesh to the mesh in MSC Apex.

>> Post-processing enhancements

Shear, moment, and tensile diagrams are supported.

Additional resources

MSC Apex 2024.1

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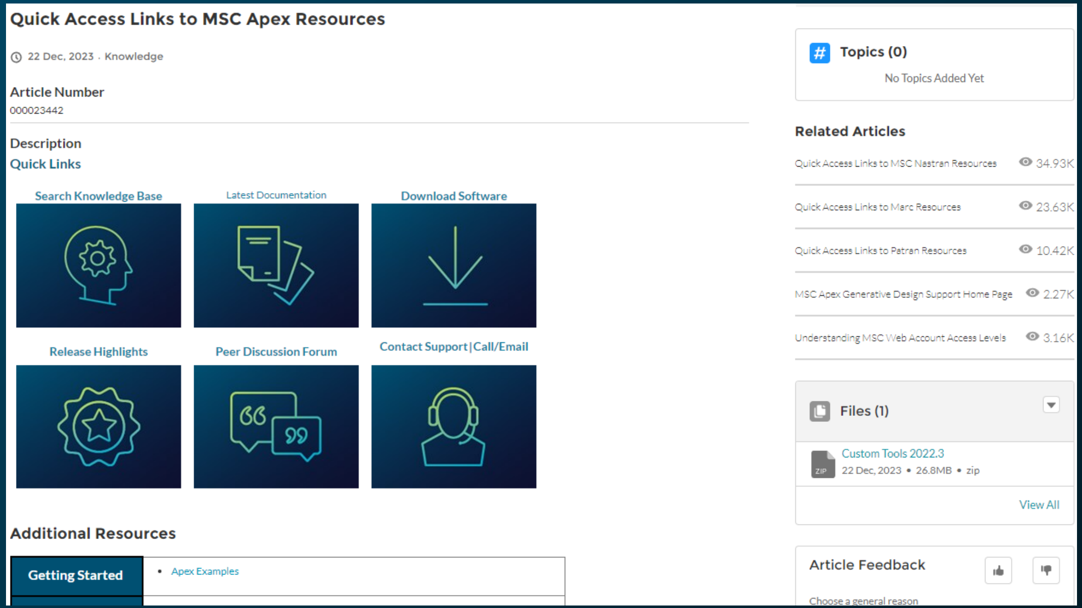
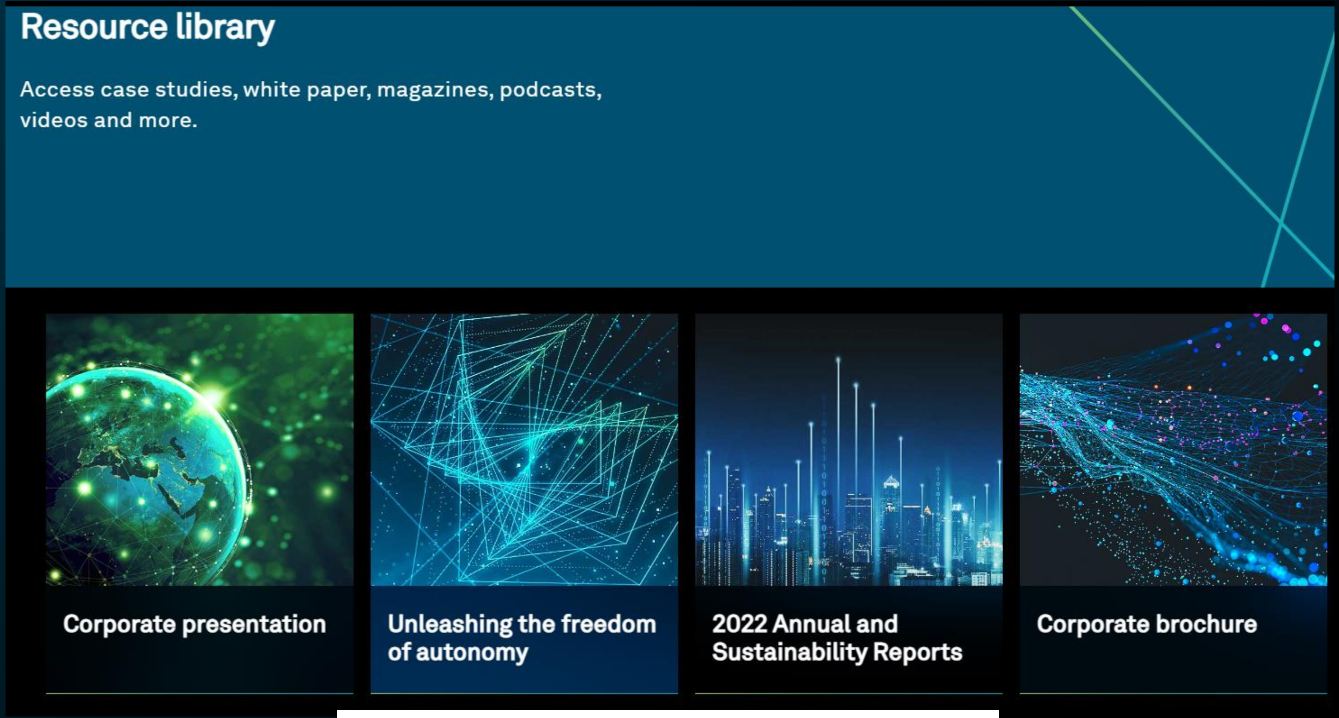
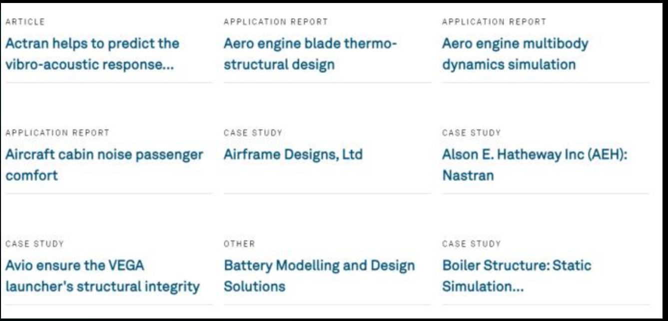
Additional Resources

Help Website

Quick Access Links to MSC Apex Resources (hexagon.com)







Support Email:
mscapex.support@hexagon.com

Licensing Questions:
msc_lic.support@hexagon.com

Report a crash:
Use the Integrated Reporting Tool (IRT)

MSC Apex Resource Library

Thank you

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Email

Name

Designation

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